

Thumbs Browser (Bundle 3.2)

Author: Francesco Savastano

Email: nwscomps@gmail.com

Website: <http://fssoft.it/delphicomps.html>

Class TThumbsBrowser

TCustomControl ← TThumbsBrowser

Declared in Unit NWSComps_ThumbsBrowser

Description:

The component's class. The browser is the container which handles the thumbnails visualization and the user interaction.

How to change Languages globally for our components:

uses NWSComps_Types; //add this unit to your uses clause

NWSCOMPS.Language := nwsLng_DE; //this changes the language to German

Properties

property AllowCustomformat_ExternalReader: boolean

Set this property to allow external reader functions to load those file types defined in the custom file formats list. To add a custom file format and its external reader function to the list use the method:
AddCustomFileFormatRead

property BackgroundColor: TColor

The background color of the browser

property Background2ndColor: TColor

The 2nd background color of the browser, used for gradient or checkboard backgrounds

property BackgroundType: TTB_Browser_BackgroundType

The background type of the browser, can have the following values:

- 1• tbbgt_SolidColor (plain color background)
- 2• tbbgt_CheckBoard (display a checkboard as background)
- 3• tbbgt_Wallpaper (uses a picture as background, specified by the

LayoutResources property)

- 4• tbbgt_GradientH (horizontal gradient)

- 5• tbbgt_GradientV (vertical gradient);

property BottomDisplayedThumbIdx: integer read-only

The index of the bottomright thumbnail in the display

property BrowsedFolders[idx: integer]: string read-only

Get the path of the folder by its index in the list of the browsed folders

property BrowsedFoldersCount: integer read-only

Get the count of the browsed folders

property BrowsingOrientation: TTB_Browser_Orientation

The orientation can be horizontal or vertical.

property BrowsingRecursively: boolean read-only

This property will let you know whether the current browsing process is a recursive browsing

property Buffer_ThumbSize: cardinal

The Maximum size in pixels (both for width and height) for the buffer of the thumbnails. The buffer is the picture that is loaded from file and resized, and it is contained in the Thumb's IEBitmap property. When the picture loaded from file has width and height greater than this value, the picture will be resized to fit in, when not the original picture will be kept.

When the value is -1, the original picture will always be kept in memory, that is the picture will not be resized.

property CheckedCount: cardinal read-only

The number of checked thumbnails

property Centered: boolean

Specify whether the displayed rows and columns of thumbnails should be always centered

property CurrentFolder: string read-only

The currently browsed folder. That is the folder to which the browser has navigated, or the last folder browsed.

property DB:TThumbsBrowser_DB

Property DBActive: boolean

1→ Enable/Disable the database for storing and reloading thumbnails property DBExternalServer: TnxServerEngine

2→ If you don't want to use the embedded nexus db server engine, but want to link the DB to an external engine, you can do so. Assign an external nexus-db server engine object to this property.
Property DBLocation: string

3→ The folder location of the thumbnails database
property DBTableName: string

4→ The table name of the thumbnails database

property DisplayMethod: TResampleFilter

The imageEn resample method used to display thumbs on screen

property DisplayRect: TRect

The coordinates of the display area of the browser

property DragDropOptions

Use this property to set the options that must regulate the dragdrop behaviour of the browser.

property DropBehavior: TthumbsBrowser_DropBehavior

1→ Define with this property which automatic behavior to adopt when there is a Drop event:

1• ddb_Files: If the objects that are dropped are files or thumbnails with valid file names, and if the Browser has a valid FolderCurrent value (not empty), the files will be moved or copied to the FolderCurrent path.

2• ddb_Thumbs: if the objects that are dropped are thumbnails, they will be copied or moved to the browser. If they are files they will be just displayed as thumbnails in the browser. No copy or move of files occurs with this

behavior.

property IS_DragSource_TB: boolean

→ Set this to define that the browser is a drag source, when using the native dragdrop between browser objects.

property IS_DropTarget_TB: boolean

→ Set this to define that the browser is a drop target, when using the native dragdrop between browser objects.

property IS_DragSource_Explorer: boolean

→ Set this to define that the browser is a drag source, when using the Windows Explorer dragdrop

property IS_DropTarget_Explorer: boolean

→ Set this to define that the browser is a drop target, when using the Windows Explorer dragdrop

property TransferMode_TB: TthumbsBrowser_DragDropTransferMode

Set this to define which kind of transfer to perform on a drop operation using Native DragDrop. Possible values are:

1. Copy (the thumbnails will be copied to the target browser)
2. Move (the thumbnails will be moved to the target browser)
3. Detect (it will be copy or move depending whether the user hold the shift key of the mouse)

property TransferMode_Exp: TthumbsBrowser_DragDropTransferMode

Set this to define which kind of transfer to perform on a drop operation using Windows Explorer dragdrop. Possible values are:

1. Copy (the files will be copied to the target explorer window)
2. Move (the files will be moved to the target explorer window)
3. Detect (it will be copy or move depending whether the user hold the shift key of the mouse)

property AddNativeDragDropTO: string

Set this to extend the native dragdrop to other classes of components. You will need to handle the `onnativedragdropEvent` to implement the behaviour for your defined classes of controls.

property DragScrollInterval: cardinal

The interval in millisecs of the scroll timer of the native dragdrop

property FileThumbs: boolean

Specify whether to show thumbnails for files.

property FileScanner_MaxTransfer: integer

Set this to define the maximum number of files to be added to the browser at a time while performing a recursive folder scanning. The adding of files is done in batches, to allow the user interface to continue its normal behaviour. The default value is typically enough for most situations.

property Filter: string

You can choose to display just those files having a certain extension: such as `*.*` or `*.jpg`. When you change the value of this property the thumbnails are immediately filtered according to the specified extension. This filter is automatically applied whenever the browser performs a new browsing operation.

1→ the Filter `*.*` will include all kinds of files.

2→ the Filter `*.*` will include all known picture and video formats.

3→ the Filter `[RAW FILES]` will display only the raw files

4→ `*.jpg` will display only jpeg files

5→ you can specify also a filter with multiple extensions such as: `*.jpg;*.gif;*.tiff;*.png`

To define your own complex custom filter, please take a look at the `FilterThumbs` method

property `FilterExclude`: string

Specify which file extensions should be excluded from the filter. The format of this string is the same as the `Filter` property.

property `Filter_AllowMultiExt`: boolean

If this property is set to true you can specify a filter with multiple extensions such as:

`*.jpg;*.png;*.bmp`

property `Filter_Updated`: boolean read-only

This property will tell you whether the `Filter` on the file type is currently updated.

property `FolderCheckBoxes`: boolean

Determines whether the thumbnails of folders should have checkboxes.

property `FolderDefault`: `TThumbsBrowserDefaultFolder`

Determines which folder should be navigated at startup.
Can have the following values:

- 1• `tbdfNone` (no folder is loaded at startup)
- 2• `tbdfDesktop`
- 3• `tbdfDrives`
- 4• `tbdfRootDir`
- 5• `tbdfMyDocuments`
- 6• `tbdfMyPictures`
- 7• `tbdfMyVideos`
- 8• `tbdfSpecified` (the folder specified by the `FolderCurrent` property will be loaded)

property `FolderCurrent`: string

Specifies which folder is the current folder, that is currently browsed.

property `FolderMonitor_Actions`: `TWatchActions`

Specify which actions should be monitored by the folder monitor.
By default the following actions are monitored:

- 1→ `waAdded`: a file was added
- 2→ `waRemoved`: a file was removed
- 3→ `waModified`: a file was modified
- 4→ `waRenamedOld`, `waRenamedNew`: a file was renamed

property `FolderMonitor_Active`: boolean

Enable / disable Folder monitor

property `FolderMonitor_TimerInterval`: cardinal

Set the interval in milliseconds after which a new event of the folder monitor shall be fired. This does not affect the performance of the monitor, it only allows the user to receive notifications after a given time. This is useful when a large number of files is modified in a short time (e.g. few milliseconds), in this case you will not get the notifications one by one at each change, but in a batch after a given interval. The default value is useful in most cases.

property FolderNavigation: boolean

Specify whether double-clicking on the thumbnail for a folder will cause the browser to navigate inside that folder.

property FolderUpNavThumb: boolean

Specify whether to show a thumbnail to navigate to the upper folder. This option is taken into account only if FolderNavigation is True

property FolderThumbs: boolean

Specify whether to show thumbnails for folders.

property InfoFormOpened: boolean

This will tell you whether the browser information form, that shows the info of the pictures, is currently opened.

property InternetOptions: TTB_Browser_InternetParams

Use this to set the parameters for an internet connection, in case the browser should retrieve pictures from a protected internet connection.

property Language: TNWSCompsLanguage

This sets the language to translate the resource strings. This affects only the current component.

1• nwsLng_EN: English

2• nwsLng_IT: Italian

3• nwsLng_ES: Spanish

4• nwsLng_DE: German

To change the language for all our components in the application use instead the following code:

uses NWSComps_Types; //add this unit to your uses clause NWSCOMPS.Language := nwsLng_DE; //this changes the language to german

property LastClickedThumb: TBrowserThumb read-only

The thumbnail which the user clicked/selected last.

property LayoutResources: TTB_GraphicResources

This contains all the graphics resources of the various layout elements of the browser, that is checkboxes, background picture, and so on.

property MaxRows: integer

Specify the maximum number of rows of thumbnails to display. This has effect only when the BrowsingOrientation is horizontal.

property MaxCols: integer

Specify the maximum number of columns of thumbnails to display. This has effect only when the BrowsingOrientation is vertical.

property MetaData_Options: TThumbsbrowser_MetaData_Options

Use this property to specify all your options for handling metadata in picture files. This will affect the handling of metadata in the browser and in the Info Form's Metadata Editor.

Property Fields_Exif → defines a list of Exif fields that should be visible and editable

Property Fields_Iptc → defines a list of Iptc fields that should be visible and editable

Property Fields_Xmp → defines a list of Xmp fields that should be visible (IE does not support writing of XMP tags to file yet (v.6.3.0))

Property Fields_Dicom → defines a list of Dicom fields that should be visible (IE does not support writing of Dicom tags to file yet(v.6.3.0))

Property Fields_Common → a Common Field is a combination of one or more fields of any of the above types ,except Dicom: Exif, IPTC, XMP. When you define a common field, the values of all the fields that make up the common field will be read and written together in synchronization. This allows the synchronization of the metadata, across Exif, Iptc and XMP. All the necessary checks for reading and writing from and to the fields is performed automatically.

Here is an example of Common Field:

{EXIF}[57|Title (Windows)][IPTC][2|5|Title]{XMP}[dc:Title|Title]

This is made of 3 fields:

- 1• the Exif field number 57, that is the Windows Title
- 2• the Iptc field defined by Records: 2 and Dset:5, that is the Iptc Title
- 3• the Xmp field defined by the name: dc:Title

When you read the common field, if a picture contains information in any of these 3 fields the value will appear in the common field, as a combination of the values of these 3 fields. If the values are the same, the field will not display the duplicate values.

For example if Windows Title = Mountain, Iptc Title = Himalaya and Xmp Title = Everest, the Common Field will display the value: 'Mountain, Himalaya, Everest'

When you save the common field back to file, all three fields will be overwritten with the value of the common field, that includes all the titles.

When you edit the common field and save it to file, the value that you assign in the common field will be replicated in all the fields. If some of the fields only allows reading but not writing, the values will not be written for those fields.

You can define your own common fields. When you do so, make sure that the fields included in a common field are present in the ExifFields, IptcFields, XmpFields lists.

You can change the order of the fields in any of the list, you can remove or add elements to the lists, from the Object Inspector.

If you choose to do so by code, then you have two possibilities:

- 1) Create first your list and then Assign the list:

```
mySl := TStringList.create;
try
mySl.Add('2|5|Title');
mySl.Add('2|120|Caption');
ThumbsBrowser1.Metadata_Options.IptcFields := mySl; //assign the fields
finally
mySl.free; //you have to free your own list end;
```

- 2) Edit directly the metadata list:

```
ThumbsBrowser1.Metadata_Options.IptcFields.clear;
ThumbsBrowser1.Metadata_Options.IptcFields.Add('2|5|Title');
ThumbsBrowser1.Metadata_Options.IptcFields.Add('2|120|Caption');
ThumbsBrowser1.Metadata_Options.Update; //in this case you need to call the update method
```

Property AutoSync_BottomTitle: TThumbsbrowser_MetaData_SyncOpType;

Property AutoSync_TopTitle: TThumbsbrowser_MetaData_SyncOpType;

Property AutoSync_Rating: TThumbsbrowser_MetaData_SyncOpType;

Property AutoSync_Keywords: TThumbsbrowser_MetaData_SyncOpType;

Set the values of these properties to handle the automatic synchronization of some of the thumbnails properties with the metadata contained in their files.

You can choose from 3 values: mdSyncOpNone, mdSyncOp_ReadOnly, mdSyncOp_ReadWrite

1→ mdSyncOpNone: with this option no automatic synchronization is performed

2→ mdSyncOp_ReadOnly: with this option, the values are read from the metadata in the files and are set in their corresponding properties in the thumbnails.

3→ mdSyncOp_ReadWrite: with this option, the values are read from the metadata in the files and are set in their corresponding properties in the thumbnails. Also when the user changes the values of the properties in the thumbnails, the new values are automatically written to the metadata in the files.

For example:

```
tb1.Metadata_Options.AutoSync_BottomTitle := mdSyncOp_ReadWrite; tb1.LoadFile('d:\test\metadata.jpg');
```

//after this a new thumbnail is added with the picture of the file. The value of the Bottom Title property will be loaded automatically out of the file's metadata because of the reading synchronization

```
tb1.LastClickedThumb.BottomTitle := 'new title'; //now the metadata in the file will contain the new title because of the writing synchronization
```

To decide which fields in the metadata must be associated with which synchronized properties you have to use the following properties:

property SyncField_BottomTitle: string property SyncField_TopTitle: string property SyncField_Rating: string property SyncField_Keywords: string

the value you assign to these fields must be in the format of a common field. For example:

```
tb1.Metadata_Options.SyncField_BottomTitle := '{EXIF}[57|Title (Windows)][IPTC][2|5|Title]{XMP}[dc:Title| Title]';
```

NOTE: the common field that you choose must be currently defined in the list of common fields.

property UseExifOrientationForThumbs: boolean

Specifies if you want the browser to change the orientation of the thumbnails automatically, when loading new files, if an exif orientation info is present in the metadata.

property MultiSelect: boolean

When true, you can perform multiple thumbs selection using the CTRL and SHIFT keys

property MultiThread: boolean

Specify whether the loading of the thumbnails should happen using multiple threads or a single thread.

property MultiThread_Pool_Count: integer

The number of concurrent threads to load the thumbnails from files.

property MultiThread_Timeout: dword

The time-out in milliseconds after which the loading threads should be considered to be dead (in case some unexpected problem occurred)

property NavMemory: boolean

This applies only if you have the FolderNavigation option set to true, or whenever you call the StartNavigation method.

Use this option when you want to keep in memory those thumbnails that were already explored while navigating through the folders. When you navigate from the current folder to another folder, the thumbnails for the current folder will not be deleted, but just hidden. In this way when you return to that folder again, the thumbnails will be instantly re-shown. In case their file content has changed the thumbnails will be reloaded again, anyway.

This option makes much faster the navigation to folders that were already explored, however all the thumbnails being kept in memory, you will need to make sure you have enough memory on the pc. Another drawback of having many hidden thumbnails is that it may cause some slow down.

Typically holding about 8000-10000 thumbnails in memory is acceptable.
You can choose the size of the cache using the NavMemoryMaxThumbs property

property NavMemoryMaxThumbs: boolean

Set the size of the Navigation Memory, that is the maximum number of thumbs that will be kept in memory for the cache.

property NColumns: cardinal read-only

The number of columns of the browser

property NRows: cardinal read-only

The number of rows of the browser

property PopupDefaultExplorer: boolean

When True, the browser will display the Explorer popup-menu, if no popup-menu is assigned

property OwnUserObjects: boolean

When True, the browser will handle the life cycle of the User Objects, that is will destroy the userObjects when their thumbnails are deleted.

property ResampleMethod: TResampleFilter

The ImageEn resample Filter used to resize and fit the pictures into the buffer size. (BufferThumbsSize)

property RotatedCount: cardinal read-only

Returns the number of thumbnails marked as to rotate.

property SelectedCount: cardinal read-only

Returns the number of selected thumbnails

property SelectedIndex: integer read-only

Returns the index of the last selected thumbnail

property ShowBottomTitle: boolean

If true, the browser will display the Bottom Title field for each thumbnail

property ShowCaptions: boolean

If true, the browser will display the caption field for each thumbnail

property ShowCheckBoxes: boolean

If true, the browser will display the check box for each thumbnail

property ShowDesignTestThumbs: boolean

If true, the browser will display at design-time some test thumbnails.

property ShowInfoButton: boolean

If true, the browser will display the info button for each thumbnail

property ShowRatingBox: boolean

If true, the browser will display a Rating box for each thumbnail

property ShowRotateButtons: boolean

If true, the browser will display the rotate buttons for each thumbnail

property ShowThumbnailHint: boolean

If true, the browser will display a hint with the info of the thumbnail on which the mouse cursor is over.

property ShowTopTitle: boolean

If true, the browser will display the Top Title field for each thumbnail

property SortType: TTB_Browser_SortType

Setting this will change the sorting of the thumbnails according to the sorting type selected

property Sort_Updated: boolean read-only

This property will tell you whether the Sorting (that you specify using the SortType) is applied to all the currently displayed thumbnails.

property StoreType: TTB_Thumb_StoreType

This property specifies in which way a picture should be stored after being loaded from file or from other source. If tbstore_FullImage is specified, the full picture will be kept in memory. If tbstore_Thumb is specified the picture will be resized and stored as a thumb (with the size specified by Buffer_ThumbSize property).

property StyleOptions: TTB_Browser_StyleOptions

TTB_Browser_StyleOptions has the following properties:

1• BrowserStyle: TTB_Browser_Style

The browser style can have the following values:

→ tbStyle_Custom (The user has customized one or more style elements) tbStyle_ThumbsH (This style set the →
→ browsing orientation to horizontal and use plain thumbnails)
→ tbStyle_ThumbsV (This style set the browsing orientation to vertical and use plain thumbnails)
→ tbStyle_DetailsH (This style set the browsing orientation to horizontal and browse by thumbnails with details to the right)
→ tbStyle_DetailsV (This style set the browsing orientation to vertical and browse by thumbnails with details to the right)
→ tbStyle_Columns (This style set the browsing orientation to vertical and browse by thumbnails with each caption representing a column to the right)

2• CaptionsOptions: TTB_Browser_Style_CaptionOptions.

→ SizePerc_HorzLayout: This is the size in percent of the caption when the layout of the thumbnail is horizontal. This size is relative to the thumbnail size. For example if the thumb has 100 pix width, and the percent is 150, the size of the caption will be: $100 \text{ pix} * 150/100 = 150 \text{ pix}$

→ Style:

Can have the following values:

capSt_Rows: the captions will be displayed by Rows aligned to the Left capSt_Cols: the captions will be displayed by Columns aligned to the Top capSt_RowsCentered: the captions will be displayed by Rows aligned to the Center

capSt_ColsCentered:: the captions will be displayed by Columns aligned to the Center TextPadding: The padding of the caption Text (intended horizontally or vertically)

3• ThemeEnabled: boolean: specify if the style look and colors should be taken from the current windows theme / VCLStyle

4• ThemeColorOptions: TTB_Browser_Style_ThemeColorOptions. Specifies which colors should be inherited from the current windows theme / VCLStyle

5• ThemeElements: TTB_Browser_Style_ThemeElements. Specifies which thumbnail elements should be inherited from the current windows theme / VCLStyle

property Thumb[idx:integer]: TThumbex read-only

Get the Thumbnail at the position idx in the list of visible thumbs.

property ThumbBottomTitle[idx:integer]: string

Get and Set the Bottom Title of the Thumbnail at the position idx in the list of visible thumbs.

property ThumbCaption[idx:integer]: string

Get and Set the General Caption of the Thumbnail at the position idx in the list of visible thumbs.

property ThumbFileName[idx:integer]: string read-only

Get the file name of the Thumbnail at the position idx in the list of visible thumbs.

property ThumbImage[idx:integer]: TIEBitmap read-only

Get the Image of the Thumbnail at the position idx in the list of visible thumbs.

property ThumbTopTitle[idx:integer]: string

Get and Set the Top Title of the Thumbnail at the position idx in the list of visible thumbs.

property ThumbUserObject[idx:integer]: TObject

Get and Set the User Object of the Thumbnail at the position idx in the list of visible thumbs.

property ThumbUserTag[idx:integer]: integer

Get and Set the User Tag of the Thumbnail at the position idx in the list of visible thumbs.

property ThumbCaption_Settings: TTB_Thumb_CaptionsSettings

These settings determine what should go into the thumbnails caption field

property ThumbDefaultChecked: boolean

This sets your preference of what should be the default checked state of any thumbnail when it is just created.

property ThumbLayoutType: TTB_Thumb_Layout_Type

With this option you can specify the layout of the thumbnail. The vertical layout places the caption at the bottom of the thumbnail, while the horizontal layout places the caption at the right of the thumbnail. To specify the width of the caption when you select the horizontal layout, use the ThumbCaptionSizePerc_HorzLayout property.

property ThumbsBackOpacity: cardinal

This is the opacity (0..255) of the thumbnail's background. The background is the bottom-level layer of the thumbnail. This is only used when you assign a background picture to the thumbnail. By default no background picture is used.

property ThumbsBackOpacitySelected: cardinal

This is the opacity (0..255) of the thumbnail's background when the thumbnail is selected. The background is the bottom-level layer of the thumbnail. This is only used when you assign a background picture to the thumbnail. By default no background picture is used.

property ThumbsBackPadding_Bottom: integer

This is the padding between the bottom of the thumbnail's background rectangle and the thumb's frame. Can be used to position the frame inside a custom thumb's background picture. By default no background picture is used.

property ThumbsBackPadding_Left: integer

This is the padding between the left of the thumbnail's background rectangle and the thumb's frame. Can be used to position the frame inside a custom thumb's background picture. By default no background picture is used.

property ThumbsBackPadding_Right: integer

This is the padding between the right of the thumbnail's background rectangle and the thumb's frame. Can be used to position the frame inside a custom thumb's background picture. By default no background picture is used.

property ThumbsBackPadding_Top: integer

This is the padding between the top of the thumbnail's background rectangle and the thumb's frame. Can be used to position the frame inside a custom thumb's background picture. By default no background picture is used.

property ThumbsBottomTitleBackColor: Tcolor

Specify with this the back color of the Bottom Title

property ThumbsBottomTitleBackSelectedColor: Tcolor

Specify with this the back color of the Bottom Title, when the thumb is selected

property ThumbsBottomTitleFontColor: Tcolor

Specify with this the font color of the Bottom Title

property ThumbsBottomTitleFontSelectedColor: Tcolor

Specify with this the font color of the Bottom Title, when the thumb is selected

property ThumbsCaptionBackColor: Tcolor

Specify with this the back color of the Caption

property ThumbsCaptionBackSelectedColor: Tcolor

Specify with this the back color of the Caption, when the thumb is selected

property ThumbsCaptionFontColor: Tcolor

Specify with this the font color of the Caption

property ThumbsCaptionFontSelectedColor: Tcolor

Specify with this the font color of the Caption, when the thumb is selected

property ThumbsCaptionOpacity: cardinal

This is the opacity (0..255) of the thumbnail's caption background.

property ThumbsCaptionOpacitySelected: cardinal

This is the opacity (0..255) of the thumbnail's caption background, when the thumb is selected

property ThumbsCaptionRoundnessPerc: cardinal

This is the roundness in percent of the thumb's caption

property ThumbsCount: cardinal

Returns the number of thumbnails currently displayed by the browser.

property ThumbsCount_Total: cardinal read-only

Returns the total number of thumbs (those visible + those invisible)

property ThumbsFrameBgColor: Tcolor

Specify with this the back color of the Thumbnails Frame

property ThumbsFrameBgOpacity: cardinal

This is the opacity (0..255) of the thumbnail's Frame Background.

property ThumbsFrameBgOpacitySelected: cardinal

This is the opacity (0..255) of the thumbnail's Frame Background, when the thumb is selected.

property ThumbsFrameBgSelectedColor: Tcolor

Specify with this the back color of the Thumbnails Frame, when the thumb is selected.

property ThumbsFrameBorderColor: Tcolor

Specify with this the border color of the Thumbnails Frame

property ThumbsFrameBorderOpacity: cardinal

This is the opacity (0..255) of the thumbnail's Frame Border.

property ThumbsFrameBorderOpacitySelected: cardinal

This is the opacity (0..255) of the thumbnail's Frame Border, when the thumb is selected.

property ThumbsFrameBorderSelectedColor: Tcolor

Specify with this the border color of the Thumbnails Frame, when the thumb is selected.

property ThumbsFramePadding_Bottom: integer

This is the padding between the bottom of the thumbnail's Frame rectangle and the content of the frame. Can be used to position the content inside a custom thumb's frame picture. By default no frame picture is used.

property ThumbsFramePadding_Left: integer

This is the padding between the left of the thumbnail's Frame rectangle and the content of the frame. Can be used to position the content inside a custom thumb's frame picture. By default no frame picture is used.

property ThumbsFramePadding_Right: integer

This is the padding between the right of the thumbnail's Frame rectangle and the content of the frame. Can be used to position the content inside a custom thumb's frame picture. By default no frame picture is used.

property ThumbsFramePadding_Top: integer

This is the padding between the top of the thumbnail's Frame rectangle and the content of the frame. Can be used to position the content inside a custom thumb's frame picture. By default no frame picture is used.

property ThumbsFrameRoundnessPerc: cardinal

This is the roundness in percent of the thumb's frame border.

property ThumbsFrameSize: cardinal

The thickness in pixel of the border of the thumbnails' frames

property ThumbSize: cardinal

The size in pixel of the thumbnails' pictures as they will appear on display. With this property you can set both width and height to the same value.

property ThumbSizeH: cardinal

The Height in pixel of the thumbnails' pictures as they will appear on display.

property ThumbSizeW: cardinal

The Width in pixel of the thumbnails' pictures as they will appear on display.

property ThumbsSpacing: cardinal

The distance in pixel between adjacent thumbnails. With this property you can set both the X and Y spacing to the same value.

property ThumbsSpacingX: cardinal

The horizontal distance in pixel between adjacent thumbnails.

property ThumbsSpacingY: cardinal

The vertical distance in pixel between adjacent thumbnails.

property ThumbsTitleOpacity: cardinal

This is the opacity (0..255) of the thumbnail's Titles Backgrounds (both top and bottom titles)

property ThumbsTitleOpacitySelected: cardinal

This is the opacity (0..255) of the thumbnail's Titles Backgrounds (both top and bottom titles), when the thumb is selected.

property ThumbsTitleRoundnessPerc: cardinal

This is the roundness in percent of the thumb's titles (both top and bottom titles) background.

property ThumbsTopTitleBackColor: Tcolor

Specify with this the background color of the Top Title

property ThumbsTopTitleBackSelectedColor: Tcolor

Specify with this the background color of the Top Title, when the thumb is selected.

property ThumbsTopTitleFontColor: Tcolor

Specify with this the font color of the Top Title

property ThumbsTopTitleFontSelectedColor: Tcolor

Specify with this the font color of the Top Title, when the thumb is selected.

property TopDisplayedThumbIdx: integer read-only

The index of the first thumbnail in the current display window

property WIA_DeviceIdx: integer read-only

The index of the WIA device to which the browser is connected. If no device it returns -1.

property WIA_Items: TList read-only

The items that have been retrieved from a WIA device, such as a photo-camera connected through a WIA connection, to be displayed in the browser.

Events

property OnAcceptExplorerDragDrop: TThumbsBrowser_DragDropAcceptDragDropEvent

This event occurs every time an explorer drag-drop operation could be initiated toward a target component. You can handle this event to accept or deny the request.

property OnAcceptNativeDragDrop: TThumbsBrowser_DragDropAcceptDragDropEvent

This event occurs every time a native drag-drop could be initiated toward a target component. You can handle this event to accept or deny the request.

property OnAllThumbsLoaded: TNotifyEvent

This event occurs when the browser has loaded all the visible thumbs, which are given always the priority over the invisible ones.

Notice that even after this event has been triggered, the browser may still be loading invisible thumbnails.

To know when the browser has finished all loading (of both visible and invisible thumbs), use the OnFinishedLoading event

property OnExplorerDragDrop: TDragDropEvent

TThumbsBrowser_ExplorerDragDropEvent = procedure(sender:TObject; Source:TObject; X, Y: integer; files: TStrings; var bHandled: boolean; var mode:TThumbsBrowser_DragDropTransferMode) of object;

This event occurs every time an explorer dragdrop event happens on the browser. You can handle this event to be notified that a drag-drop happened or to implement your own DragDrop behaviour.

Use the files parameter to get information about the list of files that have been dropped from Explorer Set bHandled := TRUE if you handle the DragDrop yourself

property OnfinishedLoading: TnotifyEvent

This event occurs when, after a StartBrowsing method has been called to start the browsing process, the browsing process has finished to load all the thumbnails with their pictures.

property OnFolderMonitorEvent: TThumbsBrowserFolderMonitorEvent

This event occurs when there have been activities in the folders currently monitored by the folder monitor.

By handling this event you can get the list of items that have triggered the event and decide what to do with them and whether you want to handle the event yourself or let the browser handle it for you.

The parameters are:

→ theItems: TthumbsBrowser_FolderMonitor_Items

This is a list whose items are of type: TthumbsBrowser_FolderMonitor_Item Each item has a FileName and an Action.

- FileName: string;
- Action: TwatchAction;

→ bHandled:boolean Set this to True if you want to handle yourself the event.

property OnInitialized: TTB_Browser_OnInitializedEvent

This event occurs after the loading process has started (i.e. by a StartBrowsing method call), when the browser has

finished to initialize all the thumbnails (without loading their pictures yet).

When the thumbnails are initialized, if they have never been loaded before, they hold only the information needed for the loading, such as the value of the `SourceFileName` property. After a thumbnail is loaded all its other information will be filled from file.

property `OnItemCustomDrawAfterDraw`: `TTB_Browser_OnItemCustomDraw`

This event occurs every time a thumbnail is drawn on the display and after all the drawing of the thumbnail has been performed. You can handle this event to draw additional elements on the thumbnail's canvas.

property `OnItemCustomDrawBottomTitle`: `TTB_Browser_OnItemCustomDrawText`

This event occurs every time the bottom title of a thumb is to be drawn. You can handle this event to customize the drawing of the bottom title.

property `OnItemCustomDrawCaption`: `TTB_Browser_OnItemCustomDrawText`

This event occurs every time the caption of a thumb is to be drawn. You can handle this event to customize the drawing of the caption.

property `OnItemCustomDrawFrame`: `TTB_Browser_OnItemCustomDraw`

This event occurs every time the frame of a thumb is to be drawn. You can handle this event to customize the drawing of the frame.

property `OnItemCustomDrawPicture`: `TTB_Browser_OnItemCustomDraw`

This event occurs every time the picture of a thumb is to be drawn. You can handle this event to customize the drawing of the picture.

property `OnItemCustomDrawThumbBg`: `TTB_Browser_OnItemCustomDraw`

This event occurs every time the background picture of a thumb is to be drawn. You can handle this event to customize the drawing of the background picture.

property `OnItemCustomDrawTopTitle`: `TTB_Browser_OnItemCustomDrawText`

This event occurs every time the top title of a thumb is to be drawn. You can handle this event to customize the drawing of the top title.

property `OnItemCustomSortCompare`: `TThumbsBrowserOnSortCompare`

This event is called when sorting the thumbnails using a custom sort mode. Specify your custom sort criteria by handling this event.

property `OnItemSearchCompare`: `TThumbsBrowserOnSearchCompare`

This event is called when filtering the thumbnails using a custom filter.

- 1→ Specify your custom filter criteria by handling this event.
- 2→ Apply the custom filter calling the `Filter_Search` method

property `OnNativeDragDrop`: `TThumbsBrowser_NativeDragDropEvent`

`TThumbsBrowser_NativeDragDropEvent = procedure(sender:TObject; Source:TObject; X, Y: integer; var bHandled: boolean; var mode:TThumbsBrowser_DragDropTransferMode ) of object;`

This event occurs every time a native drop happens on the browser. You can handle this event to be notified that a drag-drop happened, or to implement your own drag-drop behaviour.

Set `bHandled` to `TRUE` if you handle the dragdrop yourself.

property `OnNavigateFolder`: `TTB_Browser_OnNavigateFolderEvent`

This event occurs every time the browser navigates to a different folder.

property OnStartedLoading: TnotifyEvent

This event occurs the same moment a StartBrowsing method has been called to start the browsing process. In this stage the browsing has not yet proceeded.

property OnThumbAdded: TThumbsBrowserOnThumbEvent

This event occurs every time a new thumb is added to the browser.

property OnThumbBottomTitleClicked: TThumbsBrowserOnThumbClickedEvent

This event occurs when the user clicks on the thumb's bottom title.

property OnThumbCanAdd: TThumbsBrowserOnThumbCancelEvent

This event occurs before a new thumb is added to the browser. Handle this event to be able to cancel the adding of the thumbnail according to your criteria.

property OnThumbCanSelect: TThumbsBrowserOnThumbCancelEvent

This event occurs when a Thumb has been just selected. You can cancel the selection by handling the event and setting bCancel to true

property OnThumbCanDeSelect: TThumbsBrowserOnThumbCancelEvent

This event occurs when a Thumb has been just deselected. You can cancel the deselection by handling the event and setting bCancel to true

property OnThumbCaptionClicked: TThumbsBrowserOnThumbClickedEvent

This event occurs when the user clicks on the thumb's caption.

property OnThumbCheckClicked: TThumbsBrowserOnThumbClickedEvent

This event occurs when the user clicks on the thumb's checkbox. When this event is fired, the checkbox state has already been changed.

property OnThumbCheckStateChange: TThumbsBrowserOnThumbEvent

This event occurs when the checked state of a thumb has been changed.

property OnThumbCheckSpecialCase: TThumbsBrowserOnThumbEvent

This event occurs when the browser is checking if the thumb meets special condition, that is must be handled differently from other thumbnails. For example when the thumbnail is a folder, some properties may be enabled and other disabled. You can use this event to apply your own special case rules.

property OnThumbClicked: TThumbsBrowserOnThumbClickedEvent

This event occurs when the user clicks on a thumb.

property OnThumbClickedEX: TThumbsBrowserOnThumbClickedEXEvent

This event occurs when the user clicks on a thumb. This event gives more information about the click event.

property OnThumbInfoClicked: TThumbsBrowserOnThumbClickedEvent

This event occurs when the user clicks on the thumb's info button.

property OnThumbLoadDemand: TThumbsBrowserOnThumbEvent

Handle this event when you add thumbs manually with the intent to load their pictures on demand. For example, this code adds thumbs from a database (Table1) and then load them on demand, that is whenever they need to be displayed:


```

tb1.ClearThumbs;
tb1.LockUpdate; //lock update
try
Table1.First;
while not Table1.Eof do begin
with tb1.Add _a_Thumb do
    UserTag := Table1.RecNo;
Table1.Next;
end;
finally
tb1.UnlockUpdate;
end;

```

then implement the OnThumbLoadDemand:

```

procedure TMainForm.TB1ThumbLoadDemand(theThumb: TThumbEx; const Idx: Integer);
var ms:TMemoryStream;
begin

ms := TMemoryStream.Create;
try

// Go to the record of this ID
Table1.RecNo := theThumb.UserTag;

// Load the blob content into the thumb TBlobField(Table1.FieldName(FIELDNAME_PIC)).SaveToStream(ms);
ms.seek(0,0); //go to first position of stream theThumb.RetrieveFromStream(ms);

theThumb.Captions.SetCaption(Table1.FieldName(FIELDNAME_TITLE).AsString, cap_General);
theThumb.Captions.SetCaption('Rec. Nr. ' + inttostr(Table1.RecNo), cap_other1);

finally
ms.free;

end;

end;

```

property OnThumbLoadDemandAsync: TThumbsBrowser_Thumb_LoadThread_Event

Handle this event when you add thumbs manually with the intent to load their pictures on demand but in asynchronous mode. This offers the advantage of using other threads to load the thumbnails and it is useful when you need a very smooth browsing, especially if the pictures are of big dimensions.

For example, this code adds thumbs from a database (Table1) and then load them on demand asynchronously:

```

tb1.ClearThumbs;
tb1.LockUpdate; //lock update try
Table1.First;
while not Table1.Eof do
begin
with tb1.Add _a_Thumb do UserTag := Table1.RecNo;
Table1.Next;
end;
finally
tb1.UnlockUpdate;
end;

tb1.ReloadThumbs; //this starts the asynchronous loading

```

then implement the OnThumbLoadDemandAsync:

```

procedure TMainForm.TB1ThumbLoadDemandAsyinc(sender: TThread; theThumb: TThumbEx);
var ms:TMemoryStream;

```

```

begin
fCS.enter; //enter critical section: table is accessed from thread
ms := TMemoryStream.Create;
try

// Go to the record of this ID
Table1.RecNo := theThumb.UserTag;

// Load the blob content into the thumb
TBlobField(Table1.FieldByName(FIELDNAME_PIC)).SaveToStream(ms);
ms.seek(0,0); //go to first position of stream
theThumb.RetrieveFromStream(ms);

theThumb.Captions.SetCaption(Table1.FieldByName(FIELDNAME_TITLE).AsString, cap_General);
theThumb.Captions.SetCaption('Rec. Nr. ' + inttostr(Table1.RecNo), cap_other1);
finally
fCS.leave; //leave critical section ms.free;
end;
end;

```

You will then need to protect with the Critical Section every Access (both in read and write) to the table, for example:

```

fCS.enter; //enter critical section to avoid conflict with async loading
try
//inside here access the Table for example:
table1.Locate(FIELDNAME_TITLE, 'Flowers', [])

finally
fCS.leave;
end;

```

property OnThumbLoaded: TThumbsBrowserOnThumbEvent

This event occurs after the thumbnail has been loaded from file. You can use this to change some thumb property.

property OnThumbMouseOver: TThumbsBrowserOnThumbMouseOverEvent

This event occurs when the user is hovering the mouse on a thumb. You can handle this event and specify the options:

- 1→ mooShowMouseOverEffect: this option when present will enable the Mouse Over effect over the thumbnail. By default it is enabled.
- 2→ mooAllowChangeRating: this option when present will enable the possibility of changing the Rating by hovering the mouse on top of the Rating Box. By Default it is enabled.

property OnThumbRotateClicked: TThumbsBrowserOnThumbClickedEvent

This event occurs when the user clicks on one of the thumb's rotate button.

property OnThumbSelectionChange: TThumbsBrowserOnThumbEvent

This event occurs when the selected state of a thumb has been changed.

property OnThumbTopTitleClicked: TThumbsBrowserOnThumbClickedEvent

This event occurs when the user clicks on the thumb's top title.

property OnThumbVisibilityChange: TThumbsBrowserOnThumbEvent

This event occurs when the visibility state of a thumb has been changed.

property OnWIAProgress: TTB_Browser_ProgressEvent

This event occurs at each progress step during a WIA loading, started with StartBrowsing_WIA method.

Methods:

procedure AbortBrowsingRecursively

Use this method to abort the loading process, while loading recursively.

procedure AddCustomFileFormatRead (theFormat: TTB_Browser_FileFormat)

Use this method to define your own reader function for a custom file format.

function Add_a_Thumb (aUserObject: TObject = nil; const InsertIdx: integer = - 1): TBrowserThumb; overload

Use this method to manually add a thumbnail (with no picture). You can specify a user object that is linked to the thumbnail and the position at which to insert the thumbnail.

function Add_a_Thumb (filename: string; aUserObject: TObject = nil; const InsertIdx: integer = - 1; const bAsynchronousLoading: boolean = true): TBrowserThumb; overload

Use this method to manually add a thumbnail, loading the picture from file . You can specify a user object that is linked to the thumbnail and the position at which to insert the thumbnail. Also you can specify if the loading of the file should happen in asynchronous way.

function Add_a_Thumb (theIEBmp: TIEBitmap; const theCaption: string; aUserObject: TObject = nil; const InsertIdx: integer = - 1): TBrowserThumb; overload

Use this method to manually add a thumbnail, loading the picture from a TIEBitmap . You can specify a user object that is linked to the thumbnail and the position at which to insert the thumbnail. Also you can specify a caption.

function Add_a_Thumb (theBmp: TBitmap; const theCaption: string; aUserObject: TObject = nil; const InsertIdx: integer = - 1): TBrowserThumb; overload

Use this method to manually add a thumbnail, loading the picture from a TBitmap . You can specify a user object that is linked to the thumbnail and the position at which to insert the thumbnail. Also you can specify a caption.

function Add_File (filename: string; aUserObject: TObject = nil; const InsertIdx: integer = - 1; const bAsynchronousLoading: boolean = true): TBrowserThumb

Use this method to manually add a thumbnail, loading the picture from file . You can specify a user object that is linked to the thumbnail and the position at which to insert the thumbnail. Also you can specify if the loading of the file should happen in asynchronous way.

procedure Add_Files (filenames: Tstringlist; UserObjects: TList = nil; const InsertIdx: integer = - 1; const bAsynchronousLoading: boolean = true; const bGoToFirstAdded: boolean = false)

Use this method to manually add multiple thumbnails, loading the pictures from a list of files . You can specify a list of user objects that is linked to the thumbnails and the position at which to insert the group of thumbnails. Also you can specify if the loading of the file should happen in asynchronous way.

procedure CheckMarkAllThumbs (checkvalue: boolean)

Marks all the thumbnails as checked/unchecked depending from the passed parameter.

procedure CheckMarkThumb (Idx: integer; checkvalue: boolean)

Marks the thumbnail specified by its index as checked/unchecked depending from the **passed** parameter.

procedure CleanDatabase (const bNonExisting_Only, blnPath_Only: boolean)

Clean the database from thumbnails of files which no longer exist.

procedure ClearCustomFileFormatsRead

Use this method to clear the list of custom formats.

procedure ClearPathsofNonExistingThumbs

Use this method to remove from the list of browsed paths, those paths that belong only to thumbnails that no longer exist in the browser.

procedure ClearThumbs (const bClearPaths: boolean = true) overload

Delete all thumbs. If you specify that you want also to clear the paths, the browser will not keep a memory of the folders that were previously explored. If you choose not to clear the paths, then you can call a RefreshFolders method at later time to repopulate the thumbnails of those files that were in the paths previously explored.

procedure ClearThumbs(const sourcetypes: TTB_SourceTypes; const Condition: TTB_Browser_PickCondition; const bClearPaths: boolean = true) overload
Delete those thumbs that match the given conditions.

1→ SourceTypes is a set containing the kinds of thumbnails you want to delete.

If you specify an empty value it means you do not wish to specify the source type

2→ Condition specifies a condition by which to delete (if thumbnail is checked, or is selected, etc..) Set this value to ifNo_Condition if you do not want to specify a condition.

procedure ClearThumbsInPath (Pathname: string)

Delete thumbnails whose files are in the specified directory path

procedure ClearThumbs_Checked

Delete thumbnails which have checked property value set to true.

procedure ClearThumbs_NotInPaths

Delete thumbnails whose files are not in the specified directory path

procedure ClearThumbs_NotInWIA

Delete thumbnails of all files except those that have been retrieved using a WIA connection.

procedure ClearThumbs_NotInWPD

Delete thumbnails of all files except those that have been retrieved using a Windows Portable Device connection.

procedure ClearThumbs_Selected

Delete thumbnails which are currently selected.

procedure ClearThumbs_UnChecked

Delete thumbnails which have checked property value set to false.

procedure ClearThumbs_WIA

Delete thumbnails of those files that have been retrieved using a WIA connection.

procedure ClearThumbs_WPD

Delete thumbnails of those files that have been retrieved using a Windows Portable Device connection.

procedure CloseThumbsTransaction DEPRECATED since v.3.0.3 (Use UnlockUpdate instead)

To use in a Try..Finally block together with OpenThumbsTransaction.

function Delete_a_Thumb (idx: integer): boolean; overload

Delete the thumbnail specified by its index

procedure Delete_a_Thumb (theThumb: TthumbEx): boolean; overload

Delete the thumbnail specified by its reference

procedure Delete_a_Thumb_byAbsoluteldx (ID: integer): boolean; overload

Delete the thumbnail specified by its absolute Idx. This allows the deletion also of invisible thumbnails. It makes possible deletion by looping through all the thumbnails by their absolute indexes:

Example:

```
for i := tb1.ThumbsCount_Total - 1 downto 0 do //use backward loop for deletion
if tb1.Thumbat_Absoluteldx(i).TopTitle = 'delete me' then
tb1.Delete_a_Thumb_byAbsoluteldx(i);
```

procedure DeselectAllThumbs

Deselect all thumbnails

procedure DeSelectThumb (Idx: integer)

Deselect the thumbnail specified by its index

function Duplicate_a_Thumb (idx: integer; const InsertIdx: integer = - 1): TThumbEx; overload

Use this method to duplicate the thumbnail at position idx, and insert the duplicate at the position insertIdx.

function Duplicate_a_Thumb (theThumb: TThumbEx; const InsertIdx: integer = - 1):

Use this method to duplicate the thumbnail specified, and insert the duplicate at the position insertIdx.

procedure EnsureImageLoaded(idx: integer); overload;

Use this method to ensure that the thumbnail with index idx has a picture loaded. This will load the picture immediately, if the picture is not present.

procedure EnsureImageLoaded(aThumb: TThumbEX); overload;

Use this method to ensure that the specified thumbnail has a picture loaded. This will load the picture immediately, if the picture is not present.

procedure ExportToMultiBitmap(pickMode: TTB_Browser_PickCondition; mb: TIECustomMultiBitmap; bTakeOriginalPics: boolean);

This method will export the Items in the Browser (selected according to the pickMode) to a ImageEn MultiBitmap. You can choose whether to export only the thumbnails or to take the original Pictures by setting the value of the bTakeOriginalPics parameter

function ExportToMultiFileFormat(pickMode: TTB_Browser_PickCondition; mb: TIECustomMultiBitmap; bTakeOriginalPics: boolean): boolean;

This method will export the Items in the Browser (selected according to the pickMode) to a file that supports multiple images. You can choose whether to export only the thumbnails or to take the original Pictures by setting the value of the bTakeOriginalPics parameter

procedure FilterThumbs (const bGotoSelected: boolean = true)

Filter the thumbnails according to custom filter criteria.

→ the search criteria for the custom filter are specified by handling the OnItemSearchCompare event.

This search is temporary and it will be reset when you perform any new browsing operation such as calling the StartBrowsing method.

To reapply the custom filter you have to call FilterThumbs again.

It is best to call the `FilterThumbs` method only when the browser has finished the initialization of the thumbnails or when the loading process of the images is complete. In this way you are sure that the filter is consistently reapplied at each new browsing operation.

The events to use in these cases are respectively: `OnInitialized` and `OnFinishLoading`

Example:

```
tb1.OnInitialized := HandleTBInitialized;  
tb1.OnItemSearchCompare := HandleTBSearchCompare;
```

```
procedure TForm1.HandleTBInitialized(sender:TObject); begin  
    FilterThumbs(true);  
end;
```

//this will compare the first characters of the thumbs name with the chosen text //if there is match the thumb will be visible in the filter otherwise it will be hidden

```
procedure TForm1.HandleTBSearchCompare(thethumb: TThumbEx; idx: integer; var compareresult: boolean);  
var  
s: string; s1: string;  
begin
```

```
    s:= Edit_search.text;  
    if s = "" then compareresult := true  
    else  
        begin  
            if length(s)>length(thethumb.SourceFileNameShort) then  
                s1 := copy(thethumb.SourceFileNameShort, 1, length(thethumb.SourceFileNameShort)) else  
                s1 := copy(thethumb.SourceFileNameShort, 1, length(s));  
            compareresult := (comparetext(s1, s) = 0);  
        end;  
    end;
```

function FindNearest (Idx: integer; condition: TTB_Browser_PickCondition): integer

Use this method to find the nearest thumbnail to the index specified, that satisfies the conditions given.

procedure FolderMonitor_AddFolder(const theFolder:string; const bSubTree:boolean);

Use this method to manually start monitoring a folder of your choice. To enable automatic monitoring of browsed folders you can set `FolderMonitor_Active` property to `True`.

procedure FolderMonitor_RemoveFolder(const theFolder:string);

Use this method to manually stop monitoring a folder of your choice. To disable automatic monitoring for all folders you can set `FolderMonitor_Active` property to `False`. You can also `Lock / Unlock` the Folder Monitor or you can `Suspend` it for a given time.

function GetAbsoluteldx (theThumb: TThumbEx): integer

(Rarely needed) The absolute index is NOT the index of the thumbnails as they appear in the browser, although it could be the same as that. When thumbnails are created, they are assigned an index in the general list: this is the absolute Index. However making the thumbnails visible/invisible, sorting etc., will make this index differ from the actual one, that is the index of the thumbnails as they appear in the browser, which can be obtained by the most commonly used `GetThumbIdx` function.

function GetChecked (idx: integer): TThumbEx

Get the thumbnail with the given index in the list of checked thumbs.

function GetCheckedThumbIdxfromThumbIdx (idx: integer): integer

This will convert the normal thumb index into the index in the list of checked thumbs. If the thumb is not in the checked list it will return -1.

function GetRotated (idx: integer): TThumbEx

Get the thumbnail with the given index in the list of rotated thumbs.

procedure GetFileNames_Selected(filenamees: TStrings);

Obtain the list containing the file names of the selected Thumbnails

procedure GetFileNames_Checked(filenamees: TStrings);

Obtain the list containing the file names of the checked Thumbnails

function GetSelected (idx: integer): TThumbEx

Get the thumbnail with the given index in the list of selected thumbs.

function GetSelectedThumbIdxfromThumbIdx (idx: integer): integer

This will convert the normal thumb index into the index in the list of selected thumbs. If the thumb is not in the selected list it will return -1.

function GetThumbColumnbyIdx (Idx: integer): integer

Returns the column-number (0 based) of the thumbnail with the specified index.

function GetThumbIdx (theThumb: TThumbEx): integer

This is the normal thumb index, that is the index of the thumbnails as they appear in the browser

function GetThumbIdxbyCheckedIdx (idx: integer): integer

This will convert the Index in the list of checked thumbs into the normal thumb index.

function GetThumbIdxbyFileName (filename: string): integer

Returns the index of the thumbnail with the specified file name. If more thumbnails have the same file name the first found one.

function GetThumbIdxbyMousexy (x, y: integer): integer

Returns the index of the thumbnail which corresponds to the specified mouse position

function GetThumbIdxbySelectedIdx (idx: integer): integer

This will convert the Index in the list of selected thumbs into the normal thumb index.

function GetThumbRowbyIdx (Idx: integer): integer

Returns the row-number (0 based) of the thumbnail with the specified index.

**procedure GetThumbsList(thumbList:TList; sourcetypes:TTB_SourceTypes; condition:
TTB_Browser_PickCondition; blncludeInvisible: boolean = false);**
Use this method to get a list of thumbnails that match your specified conditions.

1→ The parameter sourcetypes specifies which type of thumbnail you want to retrieve in the list.
2→ The parameter condition specifies the condition on which to retrieve the thumbnails
3→ the parameter blncludeInvisible specifies whether the list should contain also thumbnails that are at the moment not visible in the browser

procedure GotoColumn (theCol: integer)

Move the scrollbar to bring the specified column in first position.

procedure GotoFileName (filename: string; select: boolean)

Select the thumbnail with the specified file name and move the scrollbar to bring the corresponding row or column in first position.

procedure GotoRow (theRow: integer)

Scroll to bring the specified row in first position.

procedure GotoThumbPosition (Idx: integer; select: boolean)

Select the thumbnail with the specified index and move the scrollbar to bring the corresponding row or column in first position.

procedure HideThumbs overload

Call this method to hide all thumbnails.

procedure HideThumbs(const sourcetypes: TTB_SourceTypes; const Condition: TTB_Browser_PickCondition) overload

Call this method to hide those thumbnails that match the given conditions:

1→ SourceTypes is a set containing the kinds of thumbnails you want to hide.

If you specify an empty value, it means you do not wish to specify the source type

2→ Condition specifies a condition by which to hide (if thumbnail is checked, or is selected, etc..) Set this value to ifNo_Condition if you do not want to specify a condition.

procedure HideThumbs(theFileNames:TStringList) overload

Call this method to hide those thumbnails that match the given file names.

function InPaths (const theFilename: string): boolean

Returns true if the file name specified has a path that is already in the list of browsed paths.

procedure InvertCheckedThumbsRange (fromthumb, tothumb: integer)

Invert the checked state of the thumbnails in the range between the two indexes specified.

procedure InvertSelectionThumbsRange (fromthumb, tothumb: integer)

Invert the selected state of the thumbnails in the range between the two indexes specified.

function IsFileExtSupported_Read (theExt: string): boolean

Returns true if the file extension is supported for reading in the browser.

function IsFileExtSupported_Write (theExt: string): boolean

Returns true if the file extension is supported for writing in the browser.

function IsUserFormat (theExt: string): boolean

Returns true if the file extension has a user defined format associated to it.

procedure LoadFile(const theFilename: string);

Loads the specified file into a new thumbnail

procedure LoadFiles(filenamees: TStringList);

Loads the specified files into new thumbnails

procedure LockFolderMonitor;

This will suspend the activity of the folder monitor, until the UnlockFolderMonitor method is called

procedure LockLayout;

Similar to LockUpdate, but more suitable when you are changing the layout properties. Use this to disable any update of the thumbnails on screen to take effect. This can be useful when you want to modify the layout properties for multiple thumbnails, without triggering any update.

Then you will need to Reenable Update by using the UnlockLayout method.

procedure LockLoading;

Use this to disable any loading process to take effect. This can be useful in certain cases when you want to avoid the browser to actually start loading the files until a certain moment.

Then you will need to Reenable loading by using the UnlockLoading method.

procedure LockUpdate;

Use this to disable any update of the thumbnails on screen to take effect. This can be useful when you want to modify certain properties for multiple thumbnails, without triggering any update.

Then you will need to Reenable Update by using the UnlockUpdate method.

procedure MarkRotatedAllThumbs (rm: TTB_Thumb_RotationMode)

Marks all thumbnails as to be rotated according to the rotation mode specified.

procedure MarkRotatedThumb (Idx: integer; rm: TTB_Thumb_RotationMode)

Marks the thumb at the specified index as to be rotated according to the rotation mode specified.

procedure Move_a_Thumb (fromIdx, Toldx: integer)

Moves a thumbnail from position fromIdx to position toldx.

procedure NavigateToFolder (theFolder: string)

Navigates to the folder specified. This will clear all the existing thumbs before navigation.

procedure NavigateToWPDFolderPath(aDeviceID: string; theFolderPath: string; objTypes: TIEWPDObjectTypes = [iewFile, iewFolder])

Navigates to the portable device folder specified as a path. This will clear all the existing thumbs before navigation. The objTypes parameter is provided to search on the device only folders, or files, or both

procedure NavigateToWPDFolder(aDeviceID: string; theFolderID: string; objTypes: TIEWPDObjectTypes = [iewFile, iewFolder])

Navigates to the portable device folder specified as a folder ID. This will clear all the existing thumbs before navigation.

The objTypes parameter is provided to search on the device only folders, or files, or both

procedure OpenThumbsTransaction DEPRECATED since v. 3.0.3 (Use LockUpdate instead)

Open a transaction during which the Display will not be updated. This can be used if you want to modify the properties of many thumbnails at once, and you want to avoid the refresh of the display for each thumbnail. It is a Try..finally block with CloseThumbsTransaction.

Most property setters of the Browser already use the transaction internally, so this may be not needed in many cases.

function RectThumb2Browser (idx: integer): TRect

Obtains the Rectangle coordinates - with reference to the Browser display - of the thumbnail specified by its index

procedure RefreshDisplay

Updates the display

procedure RefreshFiles

This is deprecated. Use RefreshFolders or ReloadFiles instead.

procedure RefreshFolders

Check the dates of the currently displayed files to detect if their thumbnails need to be reloaded. Also checks the content of the folders included in any of the thumbnails file paths and if new files are found they will be added to the thumbnails list.

procedure RefreshThumb (idx: integer; const blmmmediateDisplay: boolean = true)

Update the specified thumb to screen.

procedure ReloadFile (theThumb: TThumbEx); overload

Use this method to reload the picture and info of the specified thumb from file

procedure ReloadFile (idx: integer); overload

Use this method to reload the picture and info of the thumb at the specified index from file

procedure ReloadFile (const theFilename: string); overload

Use this method to reload the picture and info of the thumb with the specified filename from file

procedure ReLoadFiles (pickMode: TTB_Browser_PickCondition); overload

Use this method to reload the pictures and their info from file for all those thumbs that match the condition specified

procedure ReLoadFiles; overload

Use this method to reload the picture and info from file for all the thumbs whose source type is file or folder.

procedure ReloadFiles (filenames: Tstringlist); overload

Use this method to reload the pictures and info from file for all the thumbs having the file names contained in the specified list.

procedure ReloadThumbs (pickMode: TTB_Browser_PickCondition); overload

Use this method to reload the pictures for all the thumbs that match the condition specified. This includes not only files and folders but thumbs from every type of source.

procedure ReloadThumbs; overload

Use this method to reload the pictures for all the thumbs. This includes not only files and folders but thumbs from every type of source.

procedure ReloadThumbs(tl:TList); overload

Use this method to reload the pictures for all the thumbs in the specified list. This includes not only files and folders but thumbs from every type of source.

procedure Remove_File (filename: string)

Removes the thumbnails having as file name the specified value

procedure Remove_Files (filenames: Tstringlist)

Removes the thumbnails having as file names the specified values in a list

procedure ResortThumbs

Reapply the current sorting criteria to all the thumbs

procedure ReShowHiddenThumbs

This will make visible all the hidden thumbnails.

procedure Search

Deprecated. Use `FilterThumbs` instead

procedure SelectAllThumbs

Select all the thumbnails.

procedure SelectThumb (Idx: integer)

Selects the thumb specified by its index.

procedure SelectThumbbyFileName (filename: string)

Selects the thumbnail with the specified file name.

Function Set_a_Thumb(Idx: integer; filename: string; aUserObject:TObject=nil; const bAsynchronousLoading: boolean = true; theStoreType:TTB_Thumb_StoreType = tbstore_Unspecified): TBrowserThumb; overload

This method replace the content of an existing thumbnail (at index idx), loading it from a specified file.

Function Set_a_Thumb(Idx: integer; theEBmp: TIEBitmap; const theCaption: string = ""; aUserObject:TObject=nil; theStoreType:TTB_Thumb_StoreType = tbstore_Unspecified): TBrowserThumb; overload

This method replace the content of an existing thumbnail (at index idx), loading it from a specified TIEBitmap.

Function Set_a_Thumb(Idx: integer; theBmp: TBitmap; const theCaption: string = ""; aUserObject:TObject=nil; theStoreType:TTB_Thumb_StoreType = tbstore_Unspecified): TBrowserThumb; overload

This method replace the content of an existing thumbnail (at index idx), loading it from a specified TBitmap.

Function Set_a_Thumb(Idx: integer; theStream: TStream; const theCaption: string = ""; aUserObject:TObject=nil; theStoreType:TTB_Thumb_StoreType = tbstore_Unspecified): TBrowserThumb; overload

This method replace the content of an existing thumbnail (at index idx), loading it from a specified TStream.

procedure ShowInfo; overload

Opens the Info form which displays the information related to the specified thumbnail.

procedure ShowInfo (thethumb: TThumbEx); overload

Opens the Info form which displays the information related to the specified thumbnail.

procedure StartBrowsing (theFolder: string); overload

Starts the process of loading and displaying a sequence of picture files. It accepts as parameters a folder's name.

procedure StartBrowsing (theFileList: Tstringlist; const blfFolder_thenBrowseContent: boolean = true); overload

Starts the process of loading and displaying a sequence of picture files. It accepts as parameters a list of file names. You can specify, when a folder name is encountered in the list, if you intend to add that folder as a single thumbnail, or if you intend to add all the files contained in that folder.

procedure StartBrowsingRecursively (theFolder: string)

Starts the process of loading and displaying a sequence of picture files. It accepts as parameters a folder name. Subfolders are also checked recursively for files.

procedure StartBrowsing_WIA (const aDeviceldx: integer)

Starts the process of loading pictures from a wia device, such as a photcamera using a wia connection.

**procedure StartBrowsing_WPD(DevId: string = ""; theFolderID: string = "";
const bSubFolders: boolean = false; const theFilter:string = ""; objTypes: TIEWPDObjectTypes = [iewFile,
iewFolder]); overload**

Starts the process of loading from portable device.

→ DevId is the id of the device

→ theFolderId is the id in memory of the folder of the device. Use "" to browse the root folder.

→ bsubFolders specify whether you want to browse the subfolders

→ theFilter: specify a file filter, for example: '*.jpg;*.bmp'

→ objTypes: specify whether you want to browse for files, for folders or for both.

procedure StopBrowsing; overload

Stops the thumbnails loading process.

procedure SuspendFolderMonitor(ms: integer; const targetFileName: string = "");

This will suspend the activity of the folder monitor for a certain time. After that, the monitor will be automatically re-enabled.

Specify the ms parameter to indicate how many milliseconds the suspension should last.

Specify the optional targetFileName parameter to indicate on which file the suspension should take place. If no file is specified the suspension will be valid for all files.

function Thumbat (idx: integer): TThumbEx; overload

Get the thumbnail at the specified index

function Thumbat (x, y: integer): TThumbEx; overload

Get the thumbnail at the specified browser coordinates

function Thumbat (filename: string): TThumbEx; overload

Get the thumbnail with the specified file name

function Thumbat_Absoluteldx (idx: integer): TThumbEx

Get the thumbnail at the specified absolute index. (Rarely needed)

**procedure ThumbsSetColors (theBackColor, theBackSelectedcolor, theCaptionFontColor,
theCaptionFontSelectedColor, theCaptionBackColor, theCaptionBackSelectedColor, theTopTitleFontColor,
theTopTitleFontSelectedColor, theTopTitleBackColor, theTopTitleBackSelectedColor,
theBottomTitleFontColor, theBottomTitleFontSelectedColor, theBottomTitleBackColor,
theBottomTitleBackSelectedColor, theFrameBorderColor, theSelFrameBorderColor: tcolor)**

Set the colors of most the thumbnail's elements with one method call

procedure UnLockFolderMonitor;

This will reenale the activity of the folder monitor, after the LockFolderMonitor method was previously called

procedure UnlockLayout;

Reenables the Update after you have called a LockLayout method.

procedure UnlockLoading;

Reenables the Loading after you have called a LockLoading method.

procedure UnlockUpdate;

Reenables the Update after you have called a LockUpdate method.

function XBrowser2Thumb (x, idx: integer): integer

Converts the Browser's X coordinates into Thumbnail's coordinates

function XThumb2Browser (x, idx: integer): integer

Converts the Thumbnail's X coordinates into Browser's coordinates

function YBrowser2Thumb (y, idx: integer): integer

Converts the Browser's Y coordinates into Thumbnail's coordinates

function YThumb2Browser (y, idx: integer): integer

Converts the Thumbnail's Y coordinates into Browser's coordinates

Class TThumbEx

TPersistent ← **TThumb** ← **TThumbEx**

Declared in Unit NWSComps_ThumbsBrowser_Thumbs

Direct Subclasses:

TBrowserThumb

Description:

This class defines the thumbnail object used in the browser. A simple example on how to use it:

```
procedure TForm1.ChangeThumbTitle(idx:integer; newtitle:string);
begin
  Thumbsbrowser1.Thumb[idx].TopTitle := newtitle;
end;
```

Properties

property BottomTitle: string

The bottom title of the thumb. You can change this directly for each thumbnail.

property CaptionFont: tFont

The font of the caption of the thumb. The font type and size is also used for the top and bottom title. You can change this directly for each thumbnail. It is a referenced property, so you are not allowed to destroy the font after you set it, doing so would cause an access violation.

property Captions: Tstringlist

The caption texts of the thumb. You can change this directly for each thumbnail, but it is not advised to do so. If you change this property directly, you will need also to call the refreshcaptions method.

To display a custom text instead of the captions you can handle the `OnItemCustomDrawCaption` of the browser instead.

property CaptionSettings: TTB_Thumb_CaptionsSettings

Allows you to specify which information to display in the caption. You can change this directly for each thumbnail.

property Checked: boolean

By this property you can know and change the checked state of the thumb. You can change this directly for each thumbnail.

property LinkedObject: TObject

this is deprecated, it uses internally **UserObject**. Please refer to this instead.

property RotateMode: TTB_Thumb_RotationMode

By this property you can know and change the rotation state of the thumb. You can change this directly for each thumbnail.

property Selected: boolean

By this property you can know and change the selected state of the thumb. You can change this directly for each thumbnail.

property ShowSettings: TTB_Thumb_ShowSettings

Allows you to specify which elements to display in the thumb. You can change this directly for each thumbnail.

property TopTitle: string

The top title of the thumb. You can change this directly for each thumbnail.

property TotalRect: TRect read-only

By this you can know the total size of the thumbnail

property UserObject: TObject

User may link an object to the thumb

property Visible: boolean

By this property you can know and change the visible state of the thumb. You can change this directly for each thumbnail.

Inherited Properties:

Properties in TThumb:

property IEBitmap: TIEBitmap read-only

The picture displayed in the thumbnail.

property IOParams: TIOParamsVals read-only

The parameters of the file from which the thumbnail was loaded (if any)

property SourceExifFileDate: Tdatetime

The Exif date of the file from which the thumbnail was loaded (if any).

property SourceExif_XPAuthor: string

The XP Author Exif field of the file from which the thumbnail was loaded (if any). You can set this directly for each thumbnail.

property SourceExif_XPComments: string

The XP Comments Exif field of the file from which the thumbnail was loaded (if any). You can set this directly for each thumbnail.

property SourceExif_XPKeywords: string

The XP Keywords Exif field of the file from which the thumbnail was loaded (if any). You can set this directly for each thumbnail.

property SourceExif_XPTitle: string

The XP Title Exif field of the file from which the thumbnail was loaded (if any). You can set this directly for each thumbnail.

property SourceFileDate: TDateTime

The date of the file from which the thumbnail was loaded (if any)

property SourceFileExtension: string read-only

The extension of the file from which the thumbnail was loaded (if any)

property SourceFileName: string

The complete file name of the file from which the thumbnail was loaded (if any)

property SourceFileNameShort: string read-only

The short file name of the file from which the thumbnail was loaded (if any)

property SourceFilePath: string read-only

The file path of the file from which the thumbnail was loaded (if any)

property SourceFileRes: integer

The image resolution of the file from which the thumbnail was loaded (if any)

property SourceFileSize: integer

The size in bytes of the file from which the thumbnail was loaded (if any)

property SourceFileWidth: integer

The width in pixel of the image file from which the thumbnail was loaded (if any)

property SourceHasExif: boolean

Returns True if the file from which the thumbnail was loaded has exif information.

property Source_IS_WIA: boolean read-only

Returns true if a thumbnail was created by loading an image using a WIA connection

property TotalHeight: cardinal read-only

The total height of the thumbnail

property TotalWidth: cardinal read-only

The total width of the thumbnail

property UniqueID: TGUID

The unique id of the thumbnail. Although it is not readonly, Do NOT change this manually.

property UserPointer: Pointer

you may use this pointer to attach some info to the thumb

property UserTag: integer

you may use this tag to attach some info to the thumb

Methods:

procedure Assign(Source: TPersistent);

Use this method to create a complete copy of an existing thumbnail and assign it to a thumb reference.

procedure AssignInfo(Source: TPersistent; bRefresh: boolean);

Use this method to copy the info of an existing thumbnail to a target thumbnail. By Info here are intended all informations such as file name, file size, etc.. that allows to assert the origin of the thumbnail.

procedure AssignLayout(Source: TPersistent; bRefresh: boolean);

Use this method to copy the layout of an existing thumbnail to a target thumbnail. By Layout here are intended all graphical elements.

procedure ForceRefreshDimensions

This forces the thumbnail to recalculate its dimensions and therefore also the disposition of its internal elements.

function GetHintText: string

This retrieves the content of the Hint text that is displayed in the browser when the ShowThumbnailHint is enabled.

function GetRect(HitPoint: TTB_Thumb_HitRectResult): TRect

This will return the rectangle coordinates of an element in the thumbnail. For example: HitPicture will return the rectangle of the picture element

HitCheck will return the rectangle of the checkbox HitCaption will return the rectangle of the caption

function HasConsistentOrientation: boolean

This will tell you whether there is inconsistency between the orientation of the thumbnail (as loaded from exif thumbnail) and the orientation of the actual file.

procedure PaintToCanvas(cv: Tcanvas; x0, y0: integer; roughmode: boolean);

By this method you can draw the thumbnail to a canvas. When Roughmode is true the picture will be painted by using a low quality resample mode.

procedure RefreshCaptions

This will update the display of the captions. Useful in case you change the captions directly.

procedure RetrieveFromBitmap(theBitmap: Tbitmap);

Loads the picture of the thumbnail from a Tbitmap.

procedure RetrieveFromIEBitmap(theIEBitmap: TIEBitmap);

Loads the picture of the thumbnail from a TIEBitmap.

procedure RetrieveFromSourceFile(theFilename: string); overload;

Loads the picture of the thumbnail from a file.

procedure RetrieveFromStream(st: TStream);

Loads the picture of the thumbnail from a TStream.

procedure RetrieveFromUrl(url: string);

Loads the picture of the thumbnail from a url.

procedure RetrieveFromWIA(aWIA: TIEWIA; aWIAItem: TIEWIAItem);

Loads the picture of the thumbnail from a Wia Item.

Class TThumbsBrowserShellProcessor

TComponent ← TThumbsBrowserShellProcessor
Declared in Unit NWSComps_ThumbsBrowser_Shell

Description:

This component is provided to perform shell operations having a Thumbs Browser as a source of files.

To use this component together with the browser, drop it on the form together with a Tthumbsbrowser component. Then assign the value of its AttachedBrowser property to the browser you intend to use as a source of files.

TthumbsBrowser → TthumbsBrowserShellProcessor

→ The main methods of the component all start with the SH_OP_ initials.

→ Other methods and properties are provided to support the main methods.

Properties

property Aborted: boolean

This stores the value of the aborted state. You can abort the printing process, using the method AbortProcess.

property AttachedBrowser: TThumbsBrowser read-only

In order to perform shell operations, you need first to assign an attached browser component.

property Compare_Date

Reference date by which to compare the dates of the files to be processed

property Compare_DateRange_End

Reference end-date of a date-range by which to compare the dates of the files to be processed

property Compare_DateRange_Start

Reference start-date of a date-range by which to compare the dates of the files to be processed

Events

property OnClipboardChanged: TTB_Shell_ClipboardChange_Event

Event fired whenever a change in the clipboard content occurred.

Property OnRenamedFile

Event fired when a file is moved/renamed

Property OnRenamedCopiedFile

Event fired when a file is copied and renamed

Constructors:

constructor Create(AOwner: TComponent);

Destructors:

destructor Destroy;

Methods:

procedure AbortProcess

Call this method to abort the current shell operation.

function CompareDate(dt1, dt2: tdatetime; datemode: TTB_Shell_CompareDateMode): boolean

A method function to compare dates.

procedure ResetDefault_RenameParameters

This resets the rename format to default values.

procedure SetDefaultFormat_RenameFile(theValue: TTB_Shell_FileRenameFormat)

This assigns the default value of the File rename format

procedure SetDefaultFormat_RenameFolder(theValue: TTB_Shell_FolderRenameFormat)

This assigns the default value of the Folder rename format

procedure SH_OP_CopyRename(const bUseDefaultFormat: boolean; FileRenameFormat: TTB_Shell_FileRenameFormat; FolderRenameFormat: TTB_Shell_FolderRenameFormat; tofolder: string; const bAddFolderByFileType: boolean; om: TTB_Shell_PickMode; condition: TTB_Browser_PickCondition; const bChangeThumbsNames: boolean = false)

Use this method to perform a Copy / Rename process of multiple thumbnails. Specify which thumbnails should be affected by the TTB_Shell_PickMode and the TTB_Browser_PickCondition, For example, to affect only the thumbnails that are checked in the browser and whose file date is today write:

SHP1.Compare_Date := now;

SHP1.SH_OP_CopyRename(false, TBSHELLCONST_DEFRENAMEFORMAT_FILES, TBSHELLCONST_DEFRENAMEFORMAT_FOLDERS, ' d:\', false, omPickSameDate, IfChecked);

Other parameters to consider are:

→ bAddFolderByFileType: if true when copying the files a folder will be added having the name of the extension for each file, and the files will be sorted in these folders.

→ bChangeThumbsNames: if true the names of the thumbnails in the browser will be changed to reflect the new file names

procedure SH_OP_CopyRename_Checked(const bUseDefaultFormat: boolean; FileRenameFormat: TTB_Shell_FileRenameFormat; FolderRenameFormat: TTB_Shell_FolderRenameFormat; tofolder: string; const bAddFolderByFileType: boolean; const bChangeThumbsNames: boolean = false)

Use this method to perform a Copy / Rename process of all the checked thumbnails.

For example:

```
SHP1.SH_OP_CopyRename_Checked(false, TBSHELLCONST_DEFRENAMEFORMAT_FILES,
TBSHELLCONST_DEFRENAMEFORMAT_FOLDERS, 'd:\', false);
```

Other parameters to consider are:

→ bAddFolderByFileType: if true when copying the files a folder will be added having the name of the extension for each file, and the files will be sorted in these folders.

→ bChangeThumbsNames: if true the names of the thumbnails in the browser will be changed to reflect the new file names

procedure SH_OP_CopyRename_Selected(const bUseDefaultFormat: boolean; FileRenameFormat: TTB_Shell_FileRenameFormat; FolderRenameFormat: TTB_Shell_FolderRenameFormat; tofolder: string; const bAddFolderByFileType: boolean; const bChangeThumbsNames: boolean = false)

Use this method to perform a Copy / Rename process of all the selected thumbnails.

For example:

```
SHP1.SH_OP_CopyRename_Selected(false, TBSHELLCONST_DEFRENAMEFORMAT_FILES,
TBSHELLCONST_DEFRENAMEFORMAT_FOLDERS, 'd:\', false);
```

Other parameters to consider are:

→ bAddFolderByFileType: if true when copying the files a folder will be added having the name of the extension for each file, and the files will be sorted in these folders.

→ bChangeThumbsNames: if true the names of the thumbnails in the browser will be changed to reflect the new file names

procedure SH_OP_CopytoFolder(thefolder: string; theFiles: Tstringlist); overload

Use this method to perform a Copy process of all specified file names.

procedure SH_OP_CopytoFolder(thefolder: string; om: TTB_Shell_PickMode; condition: TTB_Browser_PickCondition); overload

Use this method to perform a Copy process of all the files whose thumbnails that meet the chosen PickMode and PickCondition

procedure SH_OP_CopytoFolder_Checked(thefolder: string)

Use this method to perform a Copy process of all the files of the checked thumbnails

procedure SH_OP_CopytoFolder_Selected(thefolder: string)

Use this method to perform a Copy process of all the files of the selected thumbnails

procedure SH_OP_Copy_TO_ClipBoard(theFiles: Tstringlist); overload

Use this to copy into the clipboard all specified file names.

procedure SH_OP_Copy_TO_ClipBoard(om: TTB_Shell_PickMode; condition: TTB_Browser_PickCondition); overload

Use this to copy into the clipboard all the file names of those thumbnails that meet the chosen PickMode and

PickCondition

procedure SH_OP_Copy_TO_ClipBoard_Checked

Use this to copy into the clipboard all the file names of the checked thumbnails

procedure SH_OP_Copy_TO_ClipBoard_Selected

Use this to copy into the clipboard all the file names of the selected thumbnails

procedure SH_OP_CorrectImageOrientation(om: TTB_Shell_PickMode; condition: TTB_Browser_PickCondition)

Use this to correct the orientation of the images and of the exif thumbnails when the exif orientation was set in the camera. Works only for jpeg and Tiff files. Jpeg files are transformed lossless. Applies to all those thumbnails that meet the chosen PickMode and PickCondition.

procedure SH_OP_CorrectImageOrientation_Checked

Use this to correct the orientation of the images and of their exif thumbnails when the exif orientation was set in the camera. Works only for jpeg and Tiff files. Jpeg files are transformed lossless. Applies to all checked thumbnails.

procedure SH_OP_CorrectImageOrientation_Selected

Use this to correct the orientation of the images and of their exif thumbnails when the exif orientation was set in the camera. Works only for jpeg and Tiff files. Jpeg files are transformed lossless. Applies to all selected thumbnails.

procedure SH_OP_Cut_TO_ClipBoard(theFiles: Tstringlist); overload

Use this to cut into the clipboard all the file names of the checked thumbnails

procedure SH_OP_Cut_TO_ClipBoard(om: TTB_Shell_PickMode; condition: TTB_Browser_PickCondition); overload

Use this to cut into the clipboard all the file names of the thumbnails that meet the chosen PickMode and PickCondition.

procedure SH_OP_Cut_TO_ClipBoard_Checked

Use this to cut into the clipboard all the file names of the checked thumbnails

procedure SH_OP_Cut_TO_ClipBoard_Selected

Use this to cut into the clipboard all the file names of the selected thumbnails

procedure SH_OP_Delete(theFiles: Tstringlist); overload

Use this to delete all the files whose file names are in the specified list.

procedure SH_OP_Delete(om: TTB_Shell_PickMode; condition: TTB_Browser_PickCondition); overload

Use this to delete all the files whose thumbnails meet the chosen PickMode and PickCondition.

procedure SH_OP_Delete_Checked

Use this to delete all the files of the checked thumbnails

procedure SH_OP_Delete_Selected

Use this to delete all the files of the selected thumbnails

procedure SH_OP_GetFilenameList(var fl: Tstringlist; om: TTB_Shell_PickMode; condition: TTB_Browser_PickCondition)

Get a list of filenames by specifying the thumbnail pick mode and condition

procedure SH_OP_GetFilenameList_Checked(var fl: Tstringlist)

Get a list of filenames of the checked thumbs

procedure SH_OP_GetFilenameList_Selected(var fl: Tstringlist)

Get a list of filenames of the selected thumbs

procedure SH_OP_MovetoFolder(thefolder: string; theFiles: Tstringlist); overload

Use this method to perform a Move File process of all specified file names.

procedure SH_OP_MovetoFolder(thefolder: string; om: TTB_Shell_PickMode; condition: TTB_Browser_PickCondition); overload

Use this method to perform a Move File process of all the files whose thumbnails that meet the chosen PickMode and PickCondition

procedure SH_OP_MovetoFolder_Checked(thefolder: string)

Use this method to perform a Move File process of all the files of the checked thumbnails

procedure SH_OP_MovetoFolder_Selected(thefolder: string)

Use this method to perform a Move File process of all the files of the selected thumbnails

procedure SH_OP_Paste_FromClipboardToFolder(thefolder: string)

Use this method to paste files from the clipboard into the specified folder.

procedure SH_OP_RecycletoBin(theFiles: Tstringlist); overload

Use this to recycle to bin all the files whose file names are in the specified list.

procedure SH_OP_RecycletoBin(om: TTB_Shell_PickMode; condition: TTB_Browser_PickCondition); overload

Use this to recycle to bin all the files whose thumbnails meet the chosen PickMode and PickCondition.

procedure SH_OP_RecycletoBin_Checked;

Use this to recycle to bin all the files of the checked thumbnails

procedure SH_OP_RecycletoBin_Selected;

Use this to recycle to bin all the files of the selected thumbnails

procedure SH_OP_Rename(const bUseDefaultFormat: boolean; FileRenameFormat: TTB_Shell_FileRenameFormat; FolderRenameFormat: TTB_Shell_FolderRenameFormat; tofolder: string; const bAddFolderByFileType: boolean; om: TTB_Shell_PickMode; condition: TTB_Browser_PickCondition)

Use this method to perform a Rename process of all the files whose thumbnails that meet the chosen PickMode and PickCondition.

Other parameters to consider are:

→ bAddFolderByFileType: if true when renaming the files a folder will be added having the name of the extension for each file, and the files will be arranged in these folders.

procedure SH_OP_Rename_Checked(const bUseDefaultFormat: boolean; FileRenameFormat: TTB_Shell_FileRenameFormat; FolderRenameFormat: TTB_Shell_FolderRenameFormat; tofolder: string; const bAddFolderByFileType: boolean)

Use this method to perform a Rename process of all the files of the checked thumbnails. Other parameters to

consider are:

→ `bAddFolderByFileType`: if true when renaming the files a folder will be added having the name of the extension for each file, and the files will be arranged in these folders.

procedure SH_OP_Rename_Selected(const bUseDefaultFormat: boolean; FileRenameFormat: TTB_Shell_FileRenameFormat; FolderRenameFormat: TTB_Shell_FolderRenameFormat; tofolder: string; const bAddFolderByFileType: boolean)

Use this method to perform a Rename process of all the files of the selected thumbnails. Other parameters to consider are:

→ `bAddFolderByFileType`: if true when renaming the files a folder will be added having the name of the extension for each file, and the files will be arranged in these folders.

function SH_OP_ReturnRenamedFile(const bUseDefaultFormat: boolean ; oldname, tofolder: string; idx: integer; RenameFormat: TTB_Shell_FileRenameFormat; checkexistence: boolean): string

Use this method to get the new file name of a file when you specify its rename rules and format

procedure SH_OP_Rotate(om: TTB_Shell_PickMode; condition: TTB_Browser_PickCondition)

Use this method to perform a Rotation process (lossless in case it is supported) of all the files whose thumbnails that meet the chosen PickMode and PickCondition

procedure SH_OP_Rotate_Checked

Use this method to perform a Rotation process (lossless in case it is supported) of all the files of the checked thumbnails

procedure SH_OP_Rotate_Selected

Use this method to perform a Rotation process (lossless in case it is supported) of all the files of the selected thumbnails

procedure SH_OP_SaveChecked_To(outputFolder, outputEXT: string)

Use this method to save all the files of the checked thumbnails to the desired folder and file extension

procedure SH_OP_SaveListTo(outputFolder, outputEXT: string; theFileList: Tstringlist; var ReturnedFileNames: Tstringlist); overload

Use this method to save all the files whose names are in the specified list to the desired folder and file extension. Also you can obtain a new list with the names of the renamed files.

procedure SH_OP_SaveListTo(outputFolder, outputEXT: string; theFileList: Tstringlist); overload

Use this method to save all the files whose names are in the specified list to the desired folder and file extension.

procedure SH_OP_SaveSelected_To(outputFolder, outputEXT: string)

Use this method to save all the files of the checked thumbnails to the desired folder and file extension.

procedure SH_OP_SaveTo(outputFolder, outputEXT: string; om: TTB_Shell_PickMode; condition: TTB_Browser_PickCondition); overload

Use this method to save all the files whose thumbnails meet the specified pick mode and condition to the desired folder and file extension.

procedure SH_OP_SaveTo(outputFolder, outputEXT: string; om: TTB_Shell_PickMode; condition: TTB_Browser_PickCondition; var ReturnedFileNames: Tstringlist); overload

Use this method to save all the files whose thumbnails meet the specified pick mode and condition to the desired folder and file extension. Also you can obtain a new list with the names of the renamed files.

procedure SH_OP_ShowClipboardContentInBrowser

Use this method to display all the thumbnails of those files whose names are in the clipboard.

Class TThumbsBrowserPrintProcessor

TComponent ← **TThumbsBrowserPrintProcessor**
Declared in Unit NWSComps_ThumbsBrowser_Print

Description:

This component is provided to perform the printing of the thumbnails having a Thumbs Browser as a source of files and a Shell Processor as the link to it.

Drop it on the form together with a TThumbsBrowser component and a TthumbsBrowserShellProcessor component. Then assign the value of the AttachedBrowser property of the shell processor to the browser you intend to use as a source. And finally assign the value of the AttachedShellProcessor property of the Print processor to the shell processor you intend to use.

TthumbsBrowser → **TthumbsBrowserShellProcessor** → **TThumbsBrowserPrintProcessor**

To use it just call the DoPreviews method. It will display a form from which you can define the printing preferences and then start the print.

Properties

property Aborted: boolean

This stores the value of the aborted state. You can abort the printing process, using the method AbortProcess.

property AttachedShellProcessor: TThumbsBrowserShellProcessor

You need to attach a ShellProcessor to the PrintProcessor, before being able to print.

Constructors:

constructor Create

Destructors:

destructor Destroy

Methods:

procedure AbortProcess

Call this method to abort the current printing operation

procedure DoPreviews(dpi: cardinal; pgwidth_inch, pgheight_inch: double);

Open a dialog where the user can select the printing format and launch the printing process.

procedure DrawPage(pgidx: integer; destcanvas: Tcanvas; dpi: cardinal; pgwidth_inch, pgheight_inch: double);

This method draws the specified thumbnails page onto any canvas

procedure GetPageBMP(pgidx: integer; destbmp: Tbitmap; pgwidth_pix, pgheight_pix: integer); overload;

Returns the Bitmap of the page of thumbnails. Before calling this, you will need to call the PreparePages method.

procedure GetPageBMP(pgidx: integer; destbmp: Tbitmap; dpi: cardinal; pgwidth_inch, pgheight_inch:

double); overload;

Returns the Bitmap of the page of thumbnails. Before calling this, you will need to call the PreparePages method.

procedure PreparePages(NRowssp, NColssp: cardinal; om: TTB_Shell_PickMode; condition: TTB_Browser_PickCondition);

Basing on the settings, such as the PrintOptions and the other properties, the Print Processor automatically calculates the number and layout of the thumbnails sheets to be printed or exported to files.

procedure PrintAllPages(pgwidth_inch, pgheight_inch: double);

This command prints all the thumbnails pages. Before calling this, you will need to call the PreparePages method.

procedure PrintPage(pgidx: integer; pgwidth_inch, pgheight_inch: double);

This command prints the selected thumbnails page. Before calling this, you will need to call the PreparePages method.

procedure SavePageBMPtoFile(pgidx: integer; filename: string; dpi: cardinal; pgwidth_inch, pgheight_inch: double);

This command saves the selected thumbnails page to a file. Before calling this, you will need to call the PreparePages method.

Other Types

TbrowserThumb

This is a descendant of TthumbEx, that hides many of the public methods of the ancestor.

TTB_GraphicResources

→ most important method:

procedure AssignResource(src: TIEBitmap; resType:TTB_Thumb_GraphicResourceType);

→ properties of type TIEBitmap:

BMP_Bg, BMP_ThumbBg, BMP_ThumbBgSelected, BMP_ThumbFrame: tIEbitmap,
BMP_ThumbFrameSelected, BMP_ThumbCaptionBg, BMP_ThumbCaptionBgSelected,
BMP_Checked, BMP_UnChecked, BMP_Info, BMP_FolderUpLevel,
BMP_Rotatebtns_Strip, BMP_RotRight_Up, BMP_RotRight_Down,
BMP_RotLeft_Up, BMP_RotLeft_Down

→ properties of type boolean:

HAS_Bg, HAS_ThumbBg, HAS_ThumbBgSelected, HAS_ThumbFrame: tIEbitmap,
HAS_ThumbFrameSelected, HAS_ThumbCaptionBg, HAS_ThumbCaptionBgSelected,
HAS_Checked, HAS_UnChecked, HAS_Info, HAS_FolderUpLevel,
HAS_Rotatebtns_Strip, HAS_RotRight_Up, HAS_RotRight_Down,
HAS_RotLeft_Up, HAS_RotLeft_Down

TTB_Browser_InternetParams

property ProxyAddress: string property ProxyUser: string property ProxyPassword: string

TTB_Browser_Orientation

TTB_Browser_Orientation = (tbo_horz, tbo_vert);

TTB_Browser_PickCondition

TTB_Browser_PickCondition = (IfSelected, IfChecked, IfNo_condition);

TTB_Browser_SortType

TTB_Browser_SortType = (stNameA, stNameD, stDateA, stDateD, stEXIFDateA, stEXIFDateD, stSizeA, stSizeD, stFolderA, stFolderD, stFileTypeA, stFileTypeD, stNameNaturalA, stNameNaturalD, stFolderNaturalA, stFolderNaturalD, stCustom);

TTB_Thumb_CaptionsSettings

TTB_Thumb_CaptionsSettings = set of (cap_ShowFileName, cap_ShowDateTime, cap_ShowFileSize, cap_ShowDimensions, cap_ShowEXIFDateTime, cap_ShowEXIF_XPAuthor, cap_ShowEXIF_XPTitle, cap_ShowEXIF_XPComment, cap_ShowEXIF_XPKeywords);

TTB_Thumb_HitRectResult

TTB_Thumb_HitRectResult = (HitOutside, HitInsideGeneral, HitPicture, HitCheck, HitRotateButtonLeft, HitRotateButtonRight, HitInfoBox, HitCaption, HitTopTitle, HitBottomTitle);

TTB_Thumb_RotationMode

TTB_Thumb_RotationMode = (tRmRight, TRmNone, tRmLeft);

TTB_Thumb_ShowSettings

TTB_Thumb_ShowSettings = set of (th_ShowCaption, th_ShowCheckBox, th_ShowRotateButtons, th_ShowInfoBox, th_ShowTopTitle, th_ShowBottomTitle);

TTB_Thumb_Layout_Type

TTB_Thumb_Layout_Type = (ltHorizontal, ltVertical);

TThumbsBrowser_DragDropAcceptDragDropEvent

TThumbsBrowser_DragDropAcceptDragDropEvent = procedure(sender:TObject; var bAccept: boolean; var mode:TThumbsBrowser_DragDropTransferMode) of object;

TThumbsBrowserOnSortCompare

TThumbsBrowserOnSortCompare = procedure(theThumb1, theThumb2: tthumbex; var sortResult: integer) of object;

TThumbsBrowserOnSearchCompare

TThumbsBrowserOnSearchCompare = procedure(thethumb: tthumbex; idx: integer; var comparerresult: boolean) of object;

TThumbsBrowserOnThumbEvent

TThumbsBrowserOnThumbEvent = procedure(theThumb: TThumbEx; const ldx: integer) of object;

TThumbsBrowserOnThumbClickedEvent

TThumbsBrowserOnThumbClickedEvent = procedure(ClickedThumb: TThumbEx; const ldx: integer) of object;

TThumbsBrowserOnThumbClickedEXEvent

TThumbsBrowserOnThumbClickedEXEvent = procedure(ClickedThumb: TThumbEx; const ldx: integer; ThumbClick_x, ThumbClick_y: integer; HitRect: TTB_Thumb_HitRectResult) of object;

ThumbsBrowserOnThumbCancelEvent

TThumbsBrowserOnThumbCancelEvent = procedure(theThumb: TThumbEx; var bCancel) of object;

TTB_Browser_OnInitializedEvent

TTB_Browser_OnInitializedEvent = procedure(sender:TObject; InitType:
TTB_Browser_CurrentBrowsingMode) of object;

TTB_Browser_OnItemCustomDraw

TTB_Browser_OnItemCustomDraw = procedure(thethumb: tthumbex; idx: integer; cv: Tcanvas; cv_rect:
Trect; var Handled:boolean) of object;

TTB_Browser_OnItemCustomDrawText

TTB_Browser_OnItemCustomDrawText = procedure(thethumb: tthumbex; idx: integer; cv: Tcanvas;
cv_rect: Trect;var theText: string;var theTextColor: Tcolor;var theTextSelectedColor: TColor; var
theTextBackColor: TColor;
var theTextBackSelectedColor: TColor; var theTextStyle: TFontStyles;
var theTextStyle_Selected: TFontStyles; var theTextOpacity: cardinal;
var theTextOpacitySel: cardinal; var Handled: boolean) of object;

TTB_Browser_OnNavigateFolderEvent

TTB_Browser_OnNavigateFolderEvent = procedure(theFolder:string) of object;

TTB_Browser_ProgressEvent

TTB_Browser_ProgressEvent = procedure(sender:TObject; const Pos, minPos, maxPos: integer) of object;

TTB_Shell_PickMode

TTB_Shell_PickMode = (omPickAll, omPickSelected, omPickChecked, omPickRotated, omPickSameYear,
omPickSameMonth, omPickSameDay, omPickSameYearMonth, omPickSameDate, omPickSameDateRange,
omPickSameDatetime,

omPickSameExifYear, omPickSameExifMonth, omPickSameExifDay, omPickSameExifYearMonth,
omPickSameExifDate, omPickSameExifDateRange);

TTB_Shell_ClipboardChange_Event

TTB_Shell_ClipboardChange_Event = procedure(bHasFiles:boolean; sender:TObject) of object;

TTB_Shell_CompareDateMode

TTB_Shell_CompareDateMode = (cd_date, cd_year, cd_yearmonth);

TTB_Shell_FileRenameFormat

TTB_Shell_FileRenameFormat = record FixedStr: string;
Sep_Main: String; Sep_Date: string; Sep_Time: string; Sep_FileSubject: string; Sep_FileAuthor: string;
DateFormat: string; TimeFormat: string;
RenameOptions: TTB_Shell_FileRenameOptions; CapitalizeOptions:TTB_Shell_FileRename_Capitalization;
UseExifDate: boolean;

TTB_Shell_FolderRenameFormat

TTB_Shell_FolderRenameFormat = record FixedStr: string;
Sep_Main: String; Sep_Date: string; Sep_Time: string; Sep_Subject: string; DateFormat: string;

RenameOptions: TTB_Shell_FolderRenameOptions; UseExifDate: boolean;

AddFileTypeFolder: boolean; TypeFolder_Capitalized: boolean; end;